

that these variables are readily measurable and have real economic meaning.

While macroeconomic models offer key insight into the general state of the economy they may not sufficiently capture the most accurate correlation structure of price movement across stocks. Additionally, macroeconomic models may not do a good job capturing the covariance of price movement across stocks in “new economies” or a “shifting regime” such as the sudden arrival of the financial crisis beginning in Sep. 2008.

Ross, Roll, and Chen (1986) identified the following four macroeconomic factors as having significant explanatory power with stock return. These strong relationships still hold today and are:

1. Unanticipated changes in inflation.
2. Unanticipated changes in industrial production.
3. Unanticipated changes in the yield between high-grade and low-grade corporate bonds.
4. Unanticipated changes in the yield between long-term government bonds and t-bills. This is the slope of the term structure.

Other macroeconomic factors have also been incorporated into these models include change in interest rates, growth rates, GDP, capital investment, unemployment, oil prices, housing starts, exchange rates, etc. The parameters are determined via regression analysis using monthly data over a five-year period, e.g., 60 observations.

It is often assumed that the macroeconomic factors used in the model are uncorrelated and analysts do not make any adjustment for correlation across returns. But improvements can be made to the model following the adjustment process described above.

A k-factor macroeconomic model has the form:

$$r_i = \alpha_{i0} + \hat{b}_{i1}f_1 + \hat{b}_{i2}f_2 + \dots + \hat{b}_{ik}f_k + e_i \quad (6.40)$$

Analysts need to estimate the risk exposures b_{ik} s to these macroeconomic factors.

Cross-Sectional Multi-Factor Models

Cross-sectional models estimate stock returns from a set of variables that are specific to each company rather than through factors that are common across all stocks. Cross-sectional models use stock specific factors that